

FAN SYSTEM	DUCT SECTION / COMPONENT	MATERIAL TYPE	DUCT PRESSURE CLASS
AH-1	SUPPLY	-	+2"
AH-1	RETURN	-	-2"

DUCT TYPE	DUCT LOCATION	INSULATION R-VALUE	INSTALLED INSULATION THICKNESS
SUPPLY RETURN CONDITIONED OSA	- OUTDOORS - SPACE BETWEEN ROOF & INSULATED CEILING - IN VENTED ATTIC SPACES - IN UNCONDITIONED SPACES	R-8	2"
SUPPLY CONDITIONED OSA	- ENCLOSED IN DIRECTLY CONDITIONED SPACE	NONE REQUIRED	N/A
SUPPLY CONDITIONED OSA	- IN RETURN AIR PLENUM - ALL OTHER SPACES	R-4.2	1"
RETURN	- IN INDIRECTLY CONDITIONED SPACE (E.G. SHAFT OR CEILING CAVITY)	NONE REQUIRED	N/A
UN-CONDITIONED OSA	- ANY	NONE REQUIRED	N/A

CALGREEN MANDATORY MEASURES (2022)

COMMISSIONING (§ 5.410.2)

FOR NEW BUILDINGS 10,000 SQUARE FEET AND OVER, BUILDING COMMISSIONING SHALL BE INCLUDED IN THE DESIGN AND CONSTRUCTION PROCESSES OF THE BUILDING PROJECT TO VERIFY THAT THE BUILDING SYSTEMS AND COMPONENTS WILL MEET THE OWNER OR OWNERS REPRESENTATIVE'S PROJECT REQUIREMENTS.

- TESTING AND ADJUSTING OF SYSTEMS SHALL BE REQUIRED FOR BUILDINGS LESS THAN 10,000 SQUARE FEET.

POLLUTANT CONTROL (§ 5.504)

IF THE HVAC SYSTEM IS USED DURING CONSTRUCTION, USE RETURN AIR FILTERS WITH A MERV 9 RATING, REPLACE ALL FILTERS IMMEDIATELY PRIOR TO OCCUPANCY, OR, IF THE BUILDING IS OCCUPIED DURING ALTERATION, AT THE CONCLUSION OF CONSTRUCTION.

- AT THE TIME OF ROUGH INSTALLATION, AND DURING THE STORAGE ON THE CONSTRUCTION SITE UNTIL FINAL, ALL PARTS OF THE HEATING AND COOLING EQUIPMENT, ALL DUCT AND OTHER RELATED AIR DISTRIBUTION COMPONENT OPENINGS SHALL BE COVERED.

IN MECHANICALLY VENTILATED BUILDING, PROVIDE REGULARLY OCCUPIED AREAS OF THE BUILDING WITH AIR FILTRATION. MECHANICAL VENTILATION SHALL BE PROVIDED TO ALL MECHANICALLY VENTILATED AREAS OF THE BUILDING AND CLEARLY LABELED BY THE MANUFACTURER INDICATING THE MERV RATING.

- WHERE OUTDOOR AREAS ARE PROVIDED FOR SMOKING, PROHIBIT SMOKING WITHIN 25 FT OF BUILDING ENTRIES,
- OUTDOOR AIR INTAKES, AND OPERABLE WINDOWS AND WITHIN THE BUILDING AS ALREADY PROHIBITED BY OTHER LAWS OR ORDINANCES.

INDOOR MOISTURE CONTROL (§ 5.505)

- BUILDINGS SHALL MEET OR EXCEED THE PROVISIONS OF CALIFORNIA BUILDING CODE, CCR, TITLE 24, PART 2, SECTIONS 1203 (VENTILATION) AND CHAPTER 14 (EXTERIOR WALLS).

INDOOR AIR QUALITY (§ 5.506)

FOR MECHANICALLY OR NATURALLY VENTILATED SPACES IN BUILDINGS, MEET THE MINIMUM REQUIREMENTS OF SECTION 1203.1 OF THE CALIFORNIA ENERGY CODE, OR THE APPLICABLE LOCAL CODE, WHICHEVER IS MORE STRINGENT.

FOR BUILDINGS EQUIPPED WITH DEMAND CONTROL VENTILATION, CO2 SENSORS AND VENTILATION CONTROLS SHALL BE SPECIFIED AND INSTALLED IN ACCORDANCE WITH THE REQUIREMENTS OF THE CURRENT EDITION OF THE CALIFORNIA BUILDING CODE, CCR, TITLE 24, PART 6, SECTION 1201.104.

INDOOR AIR QUALITY (§ 5.509)

NEW GLAZING, REFLECTANCES AND LOW-GLOSS EQUIPMENT, MATERIALS THAT DO NOT CONTAIN COPPER OR MANGANESE

1. PRIOR TO MECHANICAL PERMIT FINAL, A SMOKE DETECTOR SHUT-OFF TEST WILL BE REQUIRED. SMOKE DETECTOR SHUT-OFF TEST PRIOR TO MECHANICAL FINAL, COORDINATE TESTING WITH FIRE PREVENTION, DUCT SMOKE DETECTORS SUPERVISED BY THE FIRE ALARM SYSTEM PER NFPA 72, AND DETECTION OF SMOKE IN ONE SUPPLY DUCT DETECTOR SHUTS DOWN ALL HVAC UNITS.

Equipment Efficiencies (§110.2)

MECHANICAL EQUIPMENT INSTALLED IN A BUILDING SUBJECT TO THESE REGULATIONS MUST BE CERTIFIED AS MEETING CERTAIN MINIMUM EFFICIENCY. THESE REQUIREMENTS ARE CONTAINED IN §110.2, THE AFUE, COP, EER, IPLV, COMBUSTION EFFICIENCY, AND THERMAL EFFICIENCY VALUES OF ALL EQUIPMENT MUST BE DETERMINED USING THE APPLICABLE TEST PROCEDURE SPECIFIED IN THE STANDARD.

GAS AND OIL FIRED AIR FURNACES WITH INPUT RATING ≥250,000 BTU/HR SHALL HAVE AN INTERMITTENT IGNITION OR INTERRUPTED DUTY (IOD), AND HAVE EITHER POWER VENTING OR A FLUE DAMPER.

Demand Management (§110.12, 120.1(d)(3))

BUILDINGS, OTHER THAN HEALTH CARE FACILITIES, SHALL COMPLY WITH THE APPLICABLE DEMAND RESPONSIVE CONTROL, REQUIREMENTS OF SECTION 110.12(a) THRU 110.12(e).

ALL DEMAND RESPONSIVE CONTROLS SHALL BE CERTIFIED OPEN AND 2.0A OR VIRTUAL END NODE. DEMAND RESPONSIVE CONTROLS, THERMOSTAT SETPOINTS, THERMOSTAT SCHEDULING, AND APPROX 8 (A/C) NONRESIDENTIAL HVAC SYSTEMS WITH DOCT TO THE ZONE LEVEL, SHALL ALLOW CENTRALIZED DEMAND RESP. FOR NON-CRITICAL ZONES AS FOLLOWS: INCREASE COOLING SETPOINT AND DECREASE HEATING SETPOINT BY MIN. 4°F. DEMAND RESPONSE SHALL BE REQUIRED FOR ALL ZONES SUBJECT TO RATE OF CHANGE FOR TEMPERATURE INCREASE, DECREASE AND RESET, DISABLED OR MANUAL CONTROL, BY AUTHORIZED FACILITY OPERATOR, AUTOMATIC DEMAND RESPONSE CONTROL, UNDER RECEIPT OF DEMAND RESPONSE SIGNAL.

A CRITICAL ZONE IS DEFINED AS A ZONE SERVING A PROCESS WHERE RESET OF THE ZONE TEMPERATURE DURING A DEMAND RESPONSE EVENT MUST DISRUPT THE PROCESS SUBJECT BUT NOT LIMITED TO COMPUTER DATA CENTERS, TELECOM, PRIVATE BRANCH EXCHANGE ROUTES AND LABORATORIES.

ALL OCCUPABLE SPACES IN HIGH-RISE RESIDENTIAL, HOTEL/MOTEL, AND NONRESIDENTIAL BUILDINGS OTHER THAN TYPE III SHALL BE DESIGNED IN ACCORDANCE WITH SECTION 120.1(c) TO PROVIDE MECHANICAL BALANCED VENTILATION TO ALL OCCUPABLE SPACES. MECHANICAL BALANCED VENTILATION SHALL BE PROVIDED TO ALL OCCUPABLE FACILITIES SHALL BE VENTILATED IN ACCORDANCE WITH CHAPTER 4 OF THE CALIFORNIA MECHANICAL CODE.

THE OUTDOOR AIR VENTILATION RATE AND AIR DISTRIBUTION ASSUMPTIONS MADE IN THE DESIGN OF THE VENTILATING SYSTEM SHALL BE CLEARLY IDENTIFIED ON THE BUILDING PLANS REQUIRED BY §104-103 OF TITLE 24, PART 1.

PROVIDE MIN. 2" DEPTH MERV 13 FILTRATION TO SYSTEMS SUPPLYING AIR THROUGHOUT DWELLING EXCEEDING 100FT IN LENGTH. MECHANICAL, SUPPLY-ONLY VENTILATION SYSTEMS, OR SUPPLY SIDE OF MECHANICAL BALANCED VENTILATION SYSTEMS SHALL BE DESIGNED IN ACCORDANCE WITH SECTION 120.1(c).

NONRESIDENTIAL SPACES NATURALLY VENTILATED SPACES SHALL BE DESIGNED IN ACCORDANCE WITH SECTION 120.1(c)(2) THROUGH 120.1(c)(6) AND SHALL ALSO INCLUDE MECHANICAL VENTILATION IN ACCORDANCE WITH SECTION 120.1(c)(3).

TIMES OF OCCUPANCY, THE MINIMUM RATE OF OUTDOOR AIR REQUIRED BY SECTION 120.1(c) SHALL BE SUPPLIED TO EACH SPACE AT ALL TIMES WHEN THE SPACE IS USUALLY OCCUPIED.

DEMAND VENTILATION CONTROLS COMPLYING WITH §120.1(c) SHALL BE PROVIDED FOR A SPACE WITH A DESIGN OCCUPANCY OF 100 OR MORE PER PERSON. DEMAND VENTILATION CONTROLS SHALL BE DESIGNED TO PROVIDE OUTDOOR AIR FLOW RATES GREATER THAN 3.00 CFM PER PERSON, MODULATING OSA CONTROL, OR DESIGN OUTDOOR AIRFLOW RATE GREATER THAN 3.00 CFM PER PERSON.

OCCUPANT SENSOR VENTILATION CONTROL DEVICES WHEN REQUIRED BY SECTION 120.2(a)(3) SHALL BE USED TO REDUCE THE RATE OF OUTDOOR AIR FLOW WHEN OCCUPANTS ARE NOT PRESENT, IN ACCORDANCE WITH SECTION 120.1(b)(5).

DUCTING FOR ZONAL HEATING AND COOLING UNITS WHERE A RETURN PLENUM IS USED TO DISTRIBUTE OUTDOOR AIR SHALL BE DESIGNED TO MEET THE REQUIREMENTS OF SECTION 120.1(c) AND SHALL BE DESIGNED TO MEET THE REQUIREMENTS OF SECTION 120.1(c)(3). THE OUTDOOR AIR SHALL BE DUCTED TO DISCHARGE EITHER SURFACE OF THE EXTERIOR OF THE BUILDING AT A MINIMUM OF 15 FEET OF THE UNIT, SUBSTANTIALLY TOWARD THE UNIT, AND AT A VELOCITY NOT LESS THAN 500 FEET PER MINUTE.

DESIGN AND CONTROL REQUIREMENTS FOR QUANTITIES OF OUTDOOR AIR, ALL MECHANICAL VENTILATION AND DEMAND VENTILATION SHALL BE DESIGNED TO MEET THE REQUIREMENTS OF SECTION 120.1(c) AND SECTION 120.2(a)(3) TO ALLOW OUTSIDE AIR RATES TO BE OPERATED AT THE LARGER OF (1) THE MINIMUM LEVELS SPECIFIED IN SECTION 120.1(c) AND (2) THE DEMAND VENTILATION CONTROLS. DEMAND VENTILATION CONTROLS SHALL BE DESIGNED TO PROVIDE OUTDOOR AIR FLOW RATES FOR A PROCESS, FOR CONTROL OF ODORS, OR FOR THE REMOVAL OF CONTAMINANTS WITHIN THE SPACE.

THERMOSTATIC CONTROLS FOR EACH ZONE. THE SUPPLY OF HEATING AND COOLING ENERGY TO EACH ZONE (CONDITIONING ZONE FOR HEATING) SHALL BE CONTROLLED BY AN INDIVIDUAL THERMOSTATIC CONTROL THAT RESPONDS TO TEMPERATURE WITHIN THE ZONE AND THAT MEETS ALL THE APPLICABLE REQUIREMENTS OF SECTION 120.2(b).

CRITERIA FOR ZONAL THERMOSTATIC CONTROLS: THE INDIVIDUAL THERMOSTATIC CONTROLS REQUIRED BY SECTION 120.2(b) SHALL MEET THE FOLLOWING CRITERIA:

- FOR HEATING, THE THERMOSTAT SHALL BE CAPABLE OF BEING SET LOCALLY OR REMOTELY, DOWN TO 55°F OR LOWER, OR FOR COOLING IT SHALL BE CAPABLE OF BEING SET LOCALLY OR REMOTELY, UP TO 65°F OR HIGHER.
- FOR HOTEL/MOTEL, GUEST ROOM THERMOSTATS: SHALL HAVE: NUMERIC TEMPERATURE SETPOINTS IN °F, AND SETPOINT STOPS ACCESSIBLE ONLY TO AUTHORIZED PERSONNEL. SUCH THAT OCCUPANTS CANNOT ADJUST MORE THAN 10°F ABOVE OR BELOW THE SETPOINT.
- HEAT PUMP CONTROLS: ALL HEAT PUMPS WITH SUPPLEMENTARY ELECTRIC RESISTANCE HEATERS SHALL BE INSTALLED WITH CONTROLS THAT COMPLY WITH SECTION 110.2(b).

SHUT-OFF AND RESET CONTROLS FOR SPACE-CONDITIONING SYSTEMS: SHALL BE INSTALLED WITH CONTROLS CAPABLE OF AUTOMATICALLY SHUTTING OFF THE SYSTEM DURING PERIODS OF NONUSE AND SHALL HAVE AN AUTOMATIC TIME DELAY OF 15 MINUTES TO 1 HOUR TO ALLOW FOR REHEATING OF THE SPACE. THE SYSTEM FOR UP TO 4 HOURS, AN OCCUPANCY SENSOR, OR A 4 HOUR TIMER THAT CAN BE MANUALLY OPERATED.

ALL HEATING OR COOLING SYSTEMS NOT CONTROLLED BY A CENTRAL ENERGY MANAGEMENT CONTROL SYSTEM (EMCS) SHALL HAVE A SHUT-OFF AND RESET CONTROL.

OCCUPANCY SENSING ZONE CONTROLS: WHERE THE SYSTEM PROVIDING SPACE CONDITIONING ALSO PROVIDES THE VENTILATION REQUIRED BY SECTION 120.1 AND INCLUDES AN OCCUPANCY SENSOR VENTILATION CONTROL AS REQUIRED BY SECTION 120.2(a), THE SYSTEM SHALL BE CAPABLE OF AUTOMATICALLY SHUTTING OFF THE SYSTEM WITH THE FOLLOWING: PLACE THE ZONE IN OCCUPANCY STANDBY MODE WHEN ALL ROOMS SERVED BY THE ZONE ARE UNOCCUPIED FOR 15 MINUTES. THE SYSTEM SHALL BE CAPABLE OF REHEATING SETPOINTS BY 2°F (0.5°F FOR DDC SYSTEMS) AND SHUT-OFF AIRFLOW TO THE ZONE WHEN IN DEADBAND.

DAMPERS FOR AIR SUPPLY AND EXHAUST EQUIPMENT: OUTDOOR AIR SUPPLY AND EXHAUST EQUIPMENT SHALL BE INSTALLED WITH DAMPERS THAT AUTOMATICALLY CLOSE UPON FAN SHUTDOWN.

ISOLATION AREA DEVICES: EACH SPACE-CONDITIONING SYSTEM SERVING MULTIPLE ZONES WITH A COMBINED CONDITIONED FLOOR AREA OF MORE THAN 20,000 SQUARE FEET SHALL BE DESIGNED, INSTALLED, AND CONTROLLED TO SERVE ISOLATION AREAS.

ALL NEWLY INSTALLED AIR HANDLER UNITS GREATER THAN 33.0 SCF BTUHR WITH AN AIR ECONOMIZER SHALL HAVE CONTROLS FOR ISOLATION AND DEADBAND.

DDC CONTROLS TO ZONE SHALL BE PROVIDED AS SPECIFIED BY TABLE 120.2.4. SPACE-CONDITIONING SYSTEMS WITH DDC CONTROLS SHALL HAVE OPTIMUM START/STOP TABLES.

THE PIPING FOR ALL SPACE-CONDITIONING AND SERVICE WATER-HEATING SYSTEMS WITH FLUID TEMPERATURES LISTED IN TABLE 120.3-A SHALL INSTALL THE AMOUNT OF INSULATION SPECIFIED.

ALL AIR DISTRIBUTION SYSTEM DUCTS AND PLENUMS, INCLUDING, BUT NOT LIMITED TO, BUILDING CAVITIES, MECHANICAL CLOSETS, AIR-HANDLER BOXES AND SUPPORT PLATFORMS USED AS DUCTS OR PLENUMS, SHALL BE INSTALLED, SEALED AND INSULATED TO MEET THE REQUIREMENTS OF THE CMC SECTIONS 601, 602, 603, 604, 605, AND SMACNA-008-2006 HVAC DUCT CONSTRUCTION STANDARDS METAL & FLEXIBLE. 3RD EDITION.

BEFORE AN OCCUPANCY PERMIT IS GRANTED THE CERTIFICATE OF ACCEPTANCE SHALL BE SUBMITTED TO THE ENFORCEMENT AGENCY THAT CERTIFIES THAT THE EQUIPMENT AND SYSTEMS MEET THE ACCEPTANCE REQUIREMENTS.

NEW NONRESIDENTIAL BUILDINGS $\geq 10,000$ SQ. FT. SHALL COMPLY WITH THE APPLICABLE REQUIREMENTS OF SECTIONS 120.8(a) THRU 120.8(i) IN THE BUILDING DESIGN AND CONSTRUCTION PROCESSES.

- ALL NEW BOILERS OVER 2,500,000 BTU/HR SHALL COMPLY WITH COMBUSTION AIR AND COMBUSTION AIR FAN REQUIREMENTS.

■ EACH FAN OR FAN ARRAY WITH A COMBINED MOTOR NAMEPLATE HORSEPOWER GREATER THAN 1.00 HP OR WITH A COMBINED FAN NAMEPLATE ELECTRICAL INPUT POWER GREATER THAN 0.89 kW SHALL HAVE A FAN ENERGY INDEX (FEI) OF 1.00 OR HIGHER AT FAN SYSTEM DESIGN CONDITIONS. EACH FAN AND FAN ARRAY USED FOR A VARIABLE-AIR-VOLUME SYSTEM THAT MEETS THE REQUIREMENTS OF SECTION 140.4(c)2 SHALL HAVE AN FEI OF 0.95 OR HIGHER AT FAN SYSTEM DESIGN CONDITIONS.

AS EXCEPT AS MODIFIED BY A WRITTEN CONTRACT SIGNED BY THE DESIGN-BUILDER, THE DRAWINGS, SPECIFICATIONS, AND OTHER DOCUMENTS, INCLUDING THOSE IN ELECTRONIC FORM PREPARED BY OR FOR AND/OR FURNISHED BY DESIGN-BUILDER ARE INSTRUMENTS OF SERVICE. THE DESIGN-BUILDER, DESIGN-BUILDER'S CONSULTANTS AND OTHER PROVIDERS OF PROFESSIONAL SERVICES INDIVIDUALLY SHALL RETAIN A FULLY PREPARED, UNRESTRICTED COPY OF ALL INSTRUMENTS OF SERVICE. THE INSTRUMENTS OF SERVICE ARE THE PROPERTY OF DESIGN-BUILDER AND OTHER RESERVED RIGHTS, INCLUDING COPYRIGHT IN THOSE INSTRUMENTS OF SERVICE FURNISHED BY THEM.

THE INSTRUMENTS OF SERVICE FURNISHED BY DESIGN-BUILDER ARE FOR SOLE USE BY DESIGN-BUILDER IN CONSTRUCTING THE SUBJECT PROJECT. THE INSTRUMENTS OF SERVICE ARE NOT SUITABLE FOR BIDDING OR FOR ANY OTHER PROJECT OR OTHER CONTRACTOR. THE DESIGN-BUILDER OR DESIGN-BUILDER'S CONSULTANTS SHALL NOT BE PERMITTED TO REPRODUCE OR REUSE THE INSTRUMENTS OF SERVICE TO PERFORM THE WORK DEPICTED IN THE INSTRUMENTS OF SERVICE, WITHOUT THE PRIOR WRITTEN CONSENT OF DESIGN-BUILDER. THE INSTRUMENTS OF SERVICE MAY NOT BE REPRODUCED OR USED FOR ANY PURPOSE EXCEPT FOR GENERAL REFERENCE. DESIGN-BUILDER DISCLAIMS ALL LIABILITY FOR ANY ADAPTATION, MODIFICATION, OR ALTERATION OF THE INSTRUMENTS OF SERVICE, OR FOR ANY OTHER UNAUTHORIZED USE OF THESE INSTRUMENTS OF SERVICE.

2. THESE DOCUMENTS WERE PREPARED FOR CONSTRUCTION BY DESIGN-BUILDER ONLY. DESIGN-BUILDER DEVELOPED THESE DOCUMENTS WITH ENGINEERING SOLUTIONS AND CONSTRUCTION MEANS AND METHODS UNIQUE TO THE PROJECT. THE INSTRUMENTS OF SERVICE ARE NOT TO BE REPRODUCED OR REUSED FOR OTHER BIDS OR BID OR BUILD FROM. IN ADDITION, THESE DRAWINGS WERE DEVELOPED FROM INFORMATION MADE AVAILABLE TO DESIGN-BUILDER ON THE DATE OF THE INSTRUMENTS OF SERVICE. DESIGN-BUILDER DOES NOT WARRANT THAT THE INSTRUMENTS OF SERVICE ARE COMPLETE, ACCURATE, OR FREE OF OMISSIONS.

3. REMOVABLE CEILING PANELS OR PANELS AT FACE OF ALL FIRES AND FIRE-SMOKE DAMPERS BY GENERAL CONTRACTOR (24" x 24" MIN) UNLESS OTHERWISE NOTED.

4. INCOMBUSTIBLE PLENUM ABOVE RETURN AIR CIRCULATING AIR BY GENERAL CONTRACTOR. CONCEALED BUILDING SPACES USED AS RETURN AIR PLUMBS SHALL BE IN COMPLIANCE WITH 2022 CMCC 602.2.

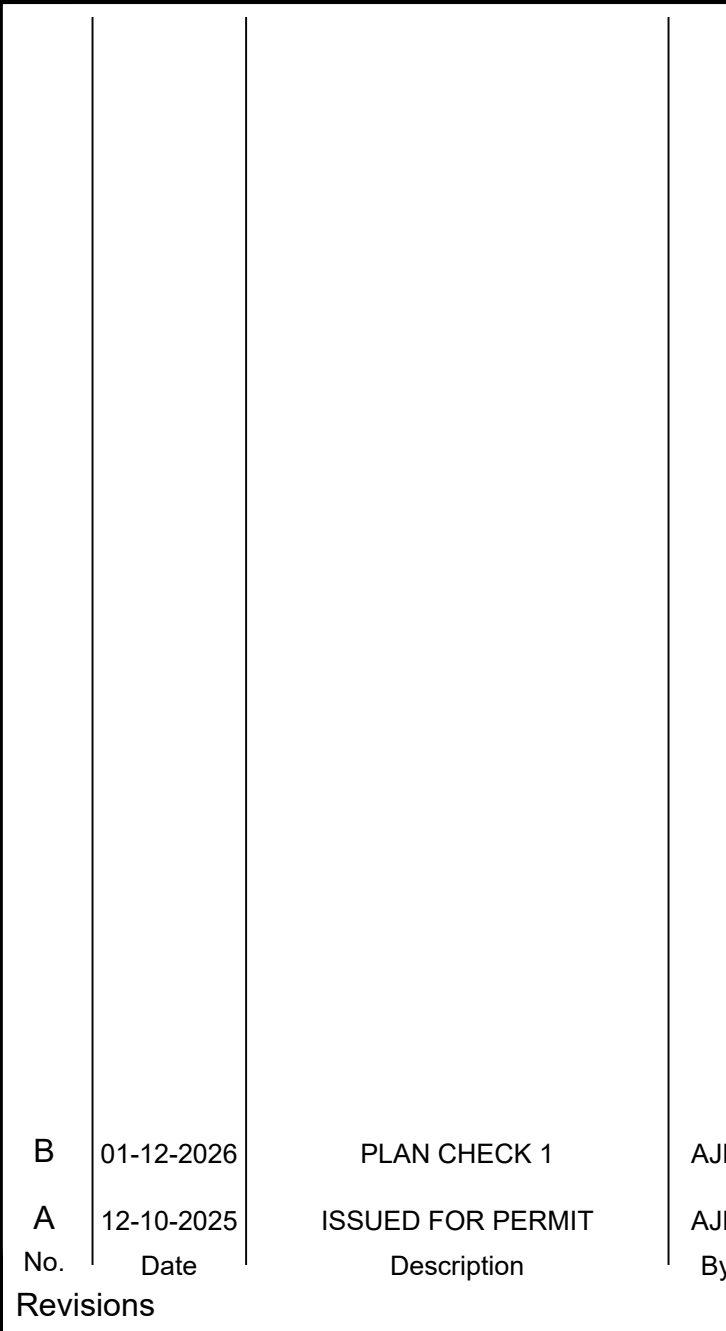
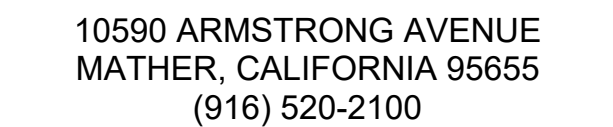
5. ACCESS DOORS AND/OR ACCESS PANELS THROUGH FIRE RATED WALLS, SHAFTS, CEILINGS, ETC., MUST EQUAL THE RATING OF THE WALLS THEY PENETRATE.

6. ALL AIR SHAFTS SHALL BE MADE AIR TIGHT BY GENERAL CONTRACTOR.

7. ALL UNDER DOORS AND DOOR LOUVERS ARE BY GENERAL CONTRACTOR.

8. ALL SPACE CONDITIONING EQUIPMENT SHALL BE LABELED AS TO WHICH AREA IT SERVES.

9. INSTALLATION OF SOFT-COOP REFRIGERANT PIPING SHALL FOLLOW THE INTERPRETATION THAT CEILING STRUCTURES AND WALL STRUCTURES MEET THE REQUIREMENTS OF HANGING PIPING REQUIRED BY 2022 CMCC 1109.7.



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